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```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score

iris = load_iris()

X = iris.data
y = iris.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = DecisionTreeClassifier()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)
```



... Accuracy: 1.0



new*



```
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2 import matplotlib.pyplot as plt
3 import seaborn as sns
4
5 from sklearn.datasets import load_iris
6 from sklearn.model_selection import
7 train_test_split
8 from sklearn.tree import
9 DecisionTreeClassifier
10
11 from sklearn.metrics import
12 accuracy_score, confusion_matrix
13 iris = load_iris()
14
15 X = iris.data
16 y = iris.target
17 X_train, X_test, y_train, y_test =
18 train_test_split(
19     X, y,
20     test_size=0.2,
21     random_state=42
22 )
23 model = DecisionTreeClassifier()
24
25 model.fit(X_train, y_train)
26 y_pred = model.predict(X_test)
27 accuracy = accuracy_score(y_test, y_pred)
28
29 print("Accuracy:", accuracy)
30 cm = confusion_matrix(y_test, y_pred)
31
32 sns.heatmap(cm, annot=True,
33             cmap="Blues")
34
```



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new*



```
train_test_split
7 from sklearn.tree import
  DecisionTreeClassifier
8 from sklearn.metrics import
  accuracy_score, confusion_matrix
9 iris = load_iris()
10
11 X = iris.data
12 y = iris.target
13 X_train, X_test, y_train, y_test =
  train_test_split(
14     X, y,
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18 model = DecisionTreeClassifier()
19
20 model.fit(X_train, y_train)
21 y_pred = model.predict(X_test)
22 accuracy = accuracy_score(y_test, y_pred)
23
24 print("Accuracy:", accuracy)
25 cm = confusion_matrix(y_test, y_pred)
26
27 sns.heatmap(cm, annot=True,
  cmap="Blues")
28
29 plt.title("Confusion Matrix")
30 plt.xlabel("Predicted")
31 plt.ylabel("Actual")
32 plt.show()
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